

① After running AWH, we get:

F_0 : Free energy of unbound state

F_1 : Free energy of bound state

$$\therefore \Delta G = F_1 - F_0$$

If we have an experimental value $\tilde{\Delta G}$

$$\mathcal{L}(\theta) = \Delta G - \tilde{\Delta G} = F_1 - F_0 - \tilde{\Delta G}$$

From AWH:

$$F_\lambda = F_\lambda^{t=1} + \Delta F_\lambda(\theta)$$

$$\mathcal{L}(\theta) = F_1^{t=1} + \Delta F_1(\theta) - F_0^{t=1} - \Delta F_0(\theta) - \tilde{\Delta G}$$

$$\mathcal{L}(\theta) = C + \Delta F_1(\theta) - \Delta F_0(\theta)$$

② We want $\frac{\partial}{\partial \theta} \mathcal{L}(\theta)$

$$\frac{\partial}{\partial \theta} \mathcal{L}(\theta) = \frac{\partial}{\partial \theta} [C + \Delta F_1(\theta) - \Delta F_0(\theta)]$$

$$= \frac{\partial}{\partial \theta} \Delta F_1(\theta) - \frac{\partial}{\partial \theta} \Delta F_0(\theta)$$

Basically we are interested in: $\frac{\partial}{\partial \theta} \Delta F_\lambda(\theta)$

$$\frac{\partial}{\partial \theta} \Delta F_\lambda(\theta) = \frac{\partial}{\partial \theta} - \ln \left[\frac{N \cdot e_\lambda + n \cdot \omega_\lambda(\theta)}{N \cdot e_\lambda + n \cdot e_\lambda} \right]$$

$$= \frac{\partial}{\partial \theta} \ln \left[N \cdot e_\lambda + n \cdot e_\lambda \right] - \frac{\partial}{\partial \theta} \ln \left[N \cdot e_\lambda + n \cdot \overbrace{\omega_\lambda(\theta)}^{u(\theta)} \right]$$

$$\frac{\partial}{\partial \theta} \ln[u(\theta)] = \frac{1}{u(\theta)} \cdot \frac{\partial}{\partial \theta} u(\theta)$$

$$\frac{\partial}{\partial \theta} u(\theta) = \frac{\partial}{\partial \theta} [N \cdot e_\lambda + n \cdot \omega_\lambda(\theta)] = n \cdot \frac{\partial}{\partial \theta} \omega_\lambda(\theta) = n \cdot \omega'_\lambda(\theta)$$

Putting everything together

$$\frac{\partial}{\partial \theta} \Delta F_\lambda(\theta) = \frac{-n \cdot \omega'_\lambda(\theta)}{N \cdot e_\lambda + n \cdot e_\lambda}$$

This means that it will only depend on the gradient of the weights w.r.t θ

$$\textcircled{3} \quad \omega_\lambda(\theta) = \frac{e_\lambda \cdot \exp[F_\lambda - \beta_\lambda \cdot u_\lambda(\theta)]}{\sum_{k \in \Lambda} e_k \cdot \exp[F_k - \beta_k \cdot u_k(\theta)]} = \frac{N}{D}$$

$$\frac{\partial}{\partial \theta} \omega_\lambda(\theta) = \frac{N' \cdot D - D' \cdot N}{D^2}$$

$$N' = \frac{\partial}{\partial \theta} [e_\lambda \cdot \exp[F_\lambda - \beta_\lambda \cdot u_\lambda(\theta)]]$$

$$= e_\lambda \cdot \beta_\lambda \cdot \exp[F_\lambda - \beta_\lambda \cdot u_\lambda(\theta)] \cdot \nabla u_\lambda(\theta)$$

$$D' = \sum_{k \in \Lambda} e_k \cdot \beta_k \cdot \exp[F_k - \beta_k \cdot u_k(\theta)] \cdot \nabla u_k(\theta)$$

This means that we "only" need to compute the gradients of the potential energy!

④ Putting all of this together is left as an exercise to the reader!